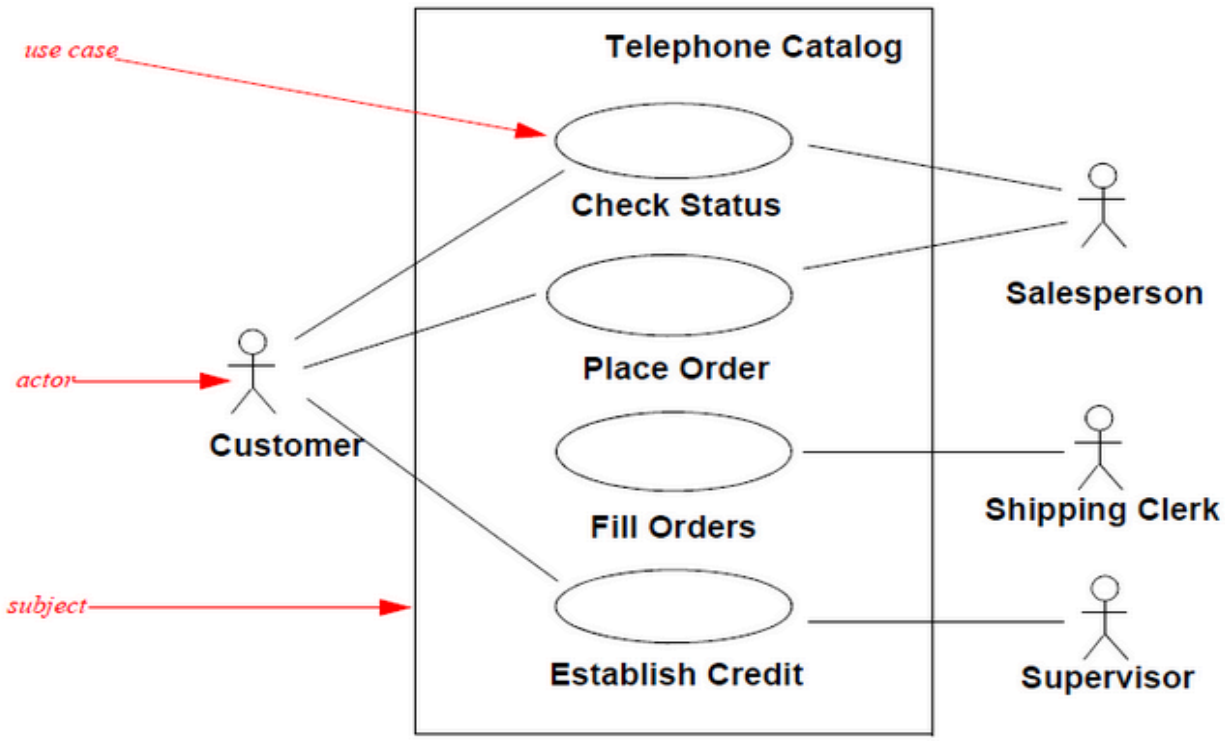
Use Case Diagrams

- A Use Case Diagram is a type of behavioral UML diagram that represents the system from an external user's perspective.
- It shows the system's functionality and the interactions between "**actors**" (external users) and "use cases" (actions the system can perform)

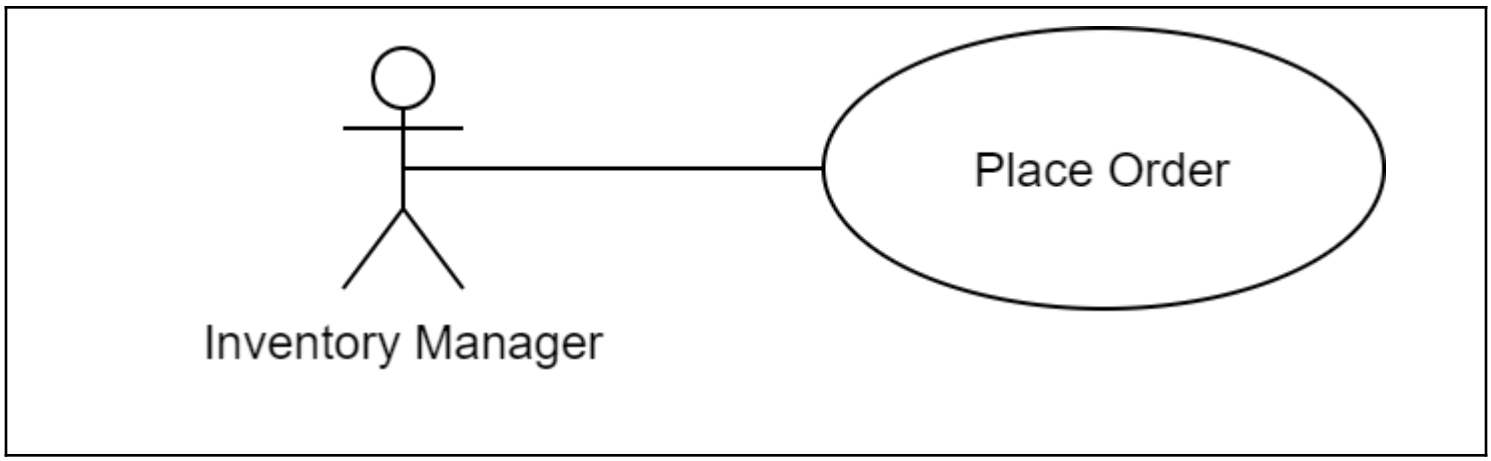


Key Elements of a Use Case Diagram

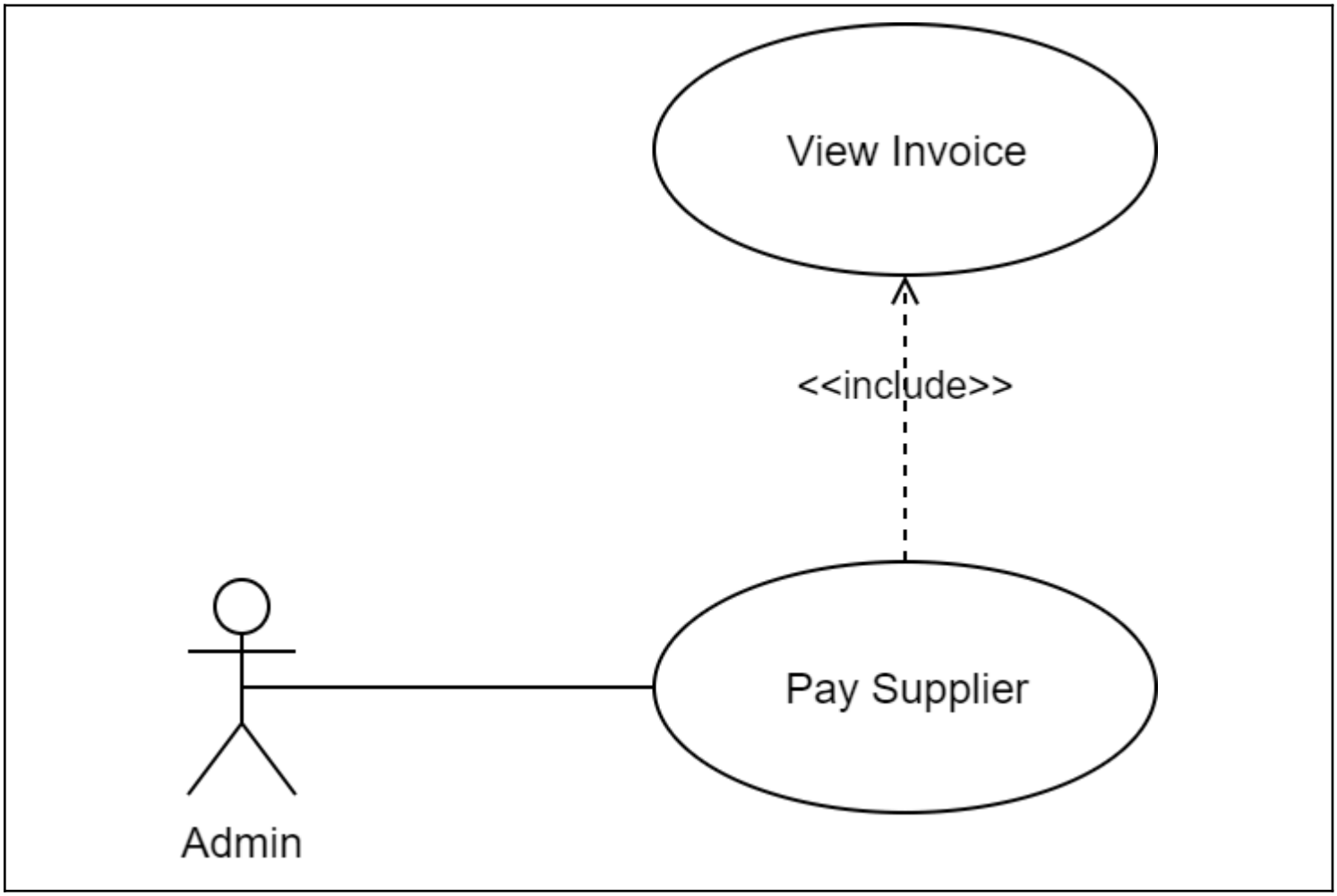
1. **System Boundary** : Defines the scope of the system. Everything inside the boundary is part of the system.
2. **Actors** : Represent entities (people, systems, devices) that interact with the system. They are usually drawn as stick figures.
3. **Use Cases** : Represent the actions or services the system provides to fulfill the actors' needs. They are usually drawn as ovals

4. **Relationships** :

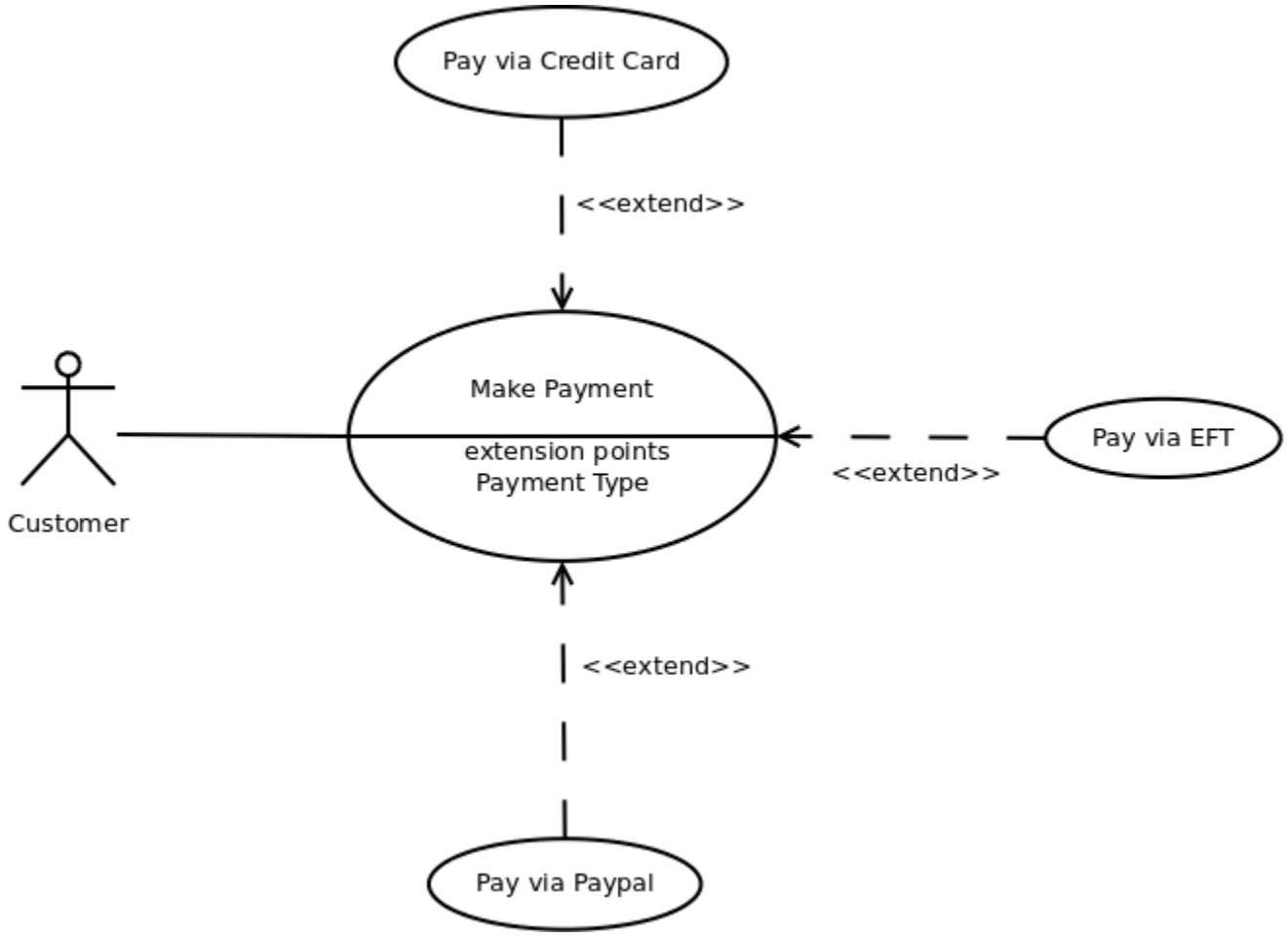
Association : A line connecting an actor to a use case, showing that the actor participates in the use case



Include : A dotted arrow pointing to a use case that is always performed as part of another use case



Extend : A dotted arrow pointing to a use case that optionally adds behavior to another use case.



Step-by-Step Example of a Use Case Diagram for a Library Management System

Step 1 : Define the System Boundary

- The system boundary is labeled as Library Management System. All features provided by the library will be inside this boundary.

Step 2 : Identify the Actors

- **Librarian** : Manages books, checks out books, and handles member registration.
- **Member** : Borrows books, returns books, and searches for books.

Step 3 : Identify the Use Cases

- **Register Member** : Used by a librarian to register a new library member.
- **Search Book** : Allows a member to search for available books.
- **Borrow Book** : Used by members to borrow a book.
- **Return Book** : Used by members to return a book.
- **Add Book** : Allows a librarian to add a new book to the library system.

Step 4 : Establish Relationships

- The Librarian has associations with Register Member and Add Book.
- The Member has associations with Search Book, Borrow Book, and Return Book.

Step 5 : Identify Include or Extend Relationships (if any)

- If Borrow Book includes checking the availability of the book before lending, we add an "Include" relationship from Borrow Book to a separate Check Availability use case.